# PAPER_787: NGC 1805 LMC Star Cluster — Three-UQFF Dense Stellar System

**Author:** Daniel T. Murphy
**Framework:** UQFF (Universal Quantum Field Superconductive Framework) — Three-UQFF Simultaneous
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## Abstract

NGC 1805 is a young open cluster in the Large Magellanic Cloud (LMC), ~163,000 light-years away. Imaged by Hubble with high resolution, NGC 1805 demonstrates the extraordinary density of young blue massive stars typical of LMC clusters. With ~$10^4$ M total cluster mass ($1.989\times10^{34}$ kg) and a half-light radius of ~3 pc ($9.46\times10^{16}$ m), it represents a compact young cluster system. Using Three-UQFF (simultaneous compressed + resonant + buoyancy modes), g_primary ≈ $1.053\times10^{-3}$ m/s² across all three modes.

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## 1. Introduction

NGC 1805 sits within the LMC's northern bar region, heavily populated with young clusters ($z \approx 0.0005$). It contains both young blue stars from recent formation and older red giants from earlier bursts, making it a mixed-age cluster. Its compact size (half-light radius ~3 pc) places it between an open cluster and a young globular cluster in terms of density. Three-UQFF probes how the compact cluster environment — much denser than galactic scales — handles the three simultaneous modes at these smaller mass/size scales.

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## 2. Three-UQFF Parameters

| Parameter | Symbol | Value | Source |
|---|---|---|---|
| Cluster mass | M | $10^4$ M = $1.989\times10^{34}$ kg | HST |
| Half-light radius | r | $9.46\times10^{16}$ m (~3 pc) | Hubble |
| SFR | — | 0 (no ongoing SF) | Quiescent now |
| Age | t | $5\times10^8$ yr = $1.578\times10^{16}$ s | Cluster age |
| M_sf | — | 0.05 | Past formation |
| Redshift | z | 0.0005 | LMC |
| v_EM | v | $10^5$ m/s | Cluster dispersion |
| B_EM | B | $10^{-5}$ T | LMC field |

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## 3. Three-UQFF Derivation

### Mode 1: Compressed UQFF

```
g_grav = 6.6743e-11 × 1.989e34 / (9.46e16)² = 1.483e-7 m/s²
H(z)×t = negligible (5e-3)
factor_sf = 1.05; factor_TRZ = 1.04
g_grav_total = 1.483e-7 × 1.05 × 1.04 = 1.619e-7 m/s²  (still << a_EM)
a_EM = 1.053e-3 m/s²
g_comp = 1.053e-3 m/s²
```

### Mode 2: Resonant UQFF

```
R_freq = 1.000285; g_res = 1.053e-3 m/s²
```

### Mode 3: Buoyancy UQFF

```
ρ_UA = 7.09e-36 kg/m³; V = (4/3)π(9.46e16)³ = 3.54e50 m³
a_Ubi << a_EM; g_buoy = 1.053e-3 m/s²
```

### Three-UQFF Simultaneous Result

```
g_compressed = 1.053e-3 m/s²
g_resonant   = 1.053e-3 m/s²
g_buoyancy   = 1.053e-3 m/s²
g_primary = 1.053e-3 m/s²
```

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## 4. Physical Interpretation

Despite being 7 orders of magnitude smaller in mass than a galaxy, NGC 1805's compact scale yields a higher classical gravitational acceleration ($1.483\times10^{-7}$ m/s$^2$) — still ~10,000× smaller than the UQFF EM term. The Three-UQFF convergence holds even at cluster scales, confirming that the electromagnetic Aether coupling dominates across all astrophysical scales from 3 pc clusters to 100 Mly galaxy groups.

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## 5. Conclusions

Three-UQFF applied to NGC 1805 LMC cluster yields g_primary ≈ $1.053\times10^{-3}$ m/s$^2$ across all three modes. UQFF scale invariance confirmed from 3 pc star clusters to 100 kly galaxy groups.

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